Question	Answer	Marks	Partial Marks
3(a)(iii)	Image at (6, 3), (6, 4), (4, 3)	**3**	**B2** for correct size and orientation but wrong position or if 3 correct points not joined **B1** for enlargement SF ½ with centre (3, 1)
3(b)	Rotation 90° [anticlockwise]oe (– 6, – 2)	**3**	**B1** for each
3(c)	Reflection $y = -x$ oe	**2**	**B1** for each
4(a)(i)	$243p^{10}$ final answer	**2**	**B1** for answer $243p^k$ or kp^{10} $(k \neq 0)$
4(a)(ii)	$9xy^4$ final answer	**2**	**B1** for answer with two correct elements in correct form of expression
4(a)(iii)	$\frac{m^2}{25}$ final answer	**1**	
4(b)	10	**4**	**B2** for $x = 8$ or for [length of rectangle =] 31 or **M1** for $5x - 9 = 3x + 7$ oe or better **M1** for $\frac{310}{(3 \times theirx + 7)}$ or $\frac{310}{(5 \times theirx - 9)}$ <u>Alt method using simultaneous eqns</u> **M1** for $5xw - 9w = 310$ **and** $3xw + 7w = 310$ **M1** for equating coefficients of xw **M1** for subtraction to eliminate term in xw
5(a)	$8^2 + 7^2 - 2 \times 7 \times 8 \times \cos 78$ oe	**M2**	**M1** for correct implicit version
	9.471.. to 9.472	**A2**	**A1** for 89.7…
5(b)	46.3 or 46.29 to 46.30…	**3**	**M2** for [sin OAC =] $\frac{7 \sin 78}{9.47}$ or **M1** for $\frac{\sin OAC}{7} = \frac{\sin 78}{9.47}$

Question	Answer	Marks	Partial Marks
5(c)	29.5 – (7 + 8 + 9.47)	**M1**	
	$\frac{360\times(29.5-(7+8+9.47))}{2\times\pi\times7}$	**M3**	**M2** for $\frac{x}{360}\times2\times\pi\times7$ = *their* arc length oe or **M1** for $\frac{x}{360}\times2\times\pi\times7$ oe
	41.15 to 41.171..	**B1**	
5(d)	45[.0] or 44.98 to 45.01 nfww	**4**	**M3** for ½ × 8 × 7 × sin 78 oe + $\frac{41.2}{360}\times\pi\times7^2$ oe OR **M1** for ½ × 8 × 7 × sin 78 oe or ½ × 8 × 9.47 × sin *their* **(b)** oe **M1** for $\frac{41.2}{360}\times\pi\times7^2$ oe
6(a)	– 2[.0], – 0.2, 2.5	**3**	**B1** for each
6(b)	Fully correct curve	**5**	**B4** for correct curve, but branches joined or **B3FT** for 9 or 10 correct plots or **B2FT** for 7 or 8 correct plots or **B1FT** for 5 or 6 correct plots and **B1 indep** two separate branches not touching or cutting *y*-axis
6(c)(i)	Correct tangent and 3 ⩽ grad ⩽ 5	**3**	**B2** for close attempt at tangent to curve at *x* = – 2 **and** answer in range OR **B1** for ruled tangent at *x* = – 2, no daylight at *x* = –2 and **M1dep** (dep on **B1** or close attempt at tangent) [at *x* = –2] for $\frac{rise}{run}$
6(c)(ii)	[*y* =] *their***(c)(i)** *x* + *their y*-intercept final answer	**2**	**Strict FT** *their y*-intercept for *their* line **M1** for *y* = *their***(c)(i)** *x* + *any value* or '*c*' oe seen or for *y* = *any value*(non-zero) *x* or '*mx*' + *their y*-intercept seen oe
6(d)(i)	1.05 to 1.25	**1**	
6(d)(ii)	– 2.3 to – 2.2 – 0.4 to – 0.3 0.3 to 0.4	**3**	**B1** for each After 0 scored **B1** for *y* = –4 ruled